NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Ephlux

TPL Trakker

Smart Asset Tracking and Predictive Maintenance System For Flight Catering

Abstract

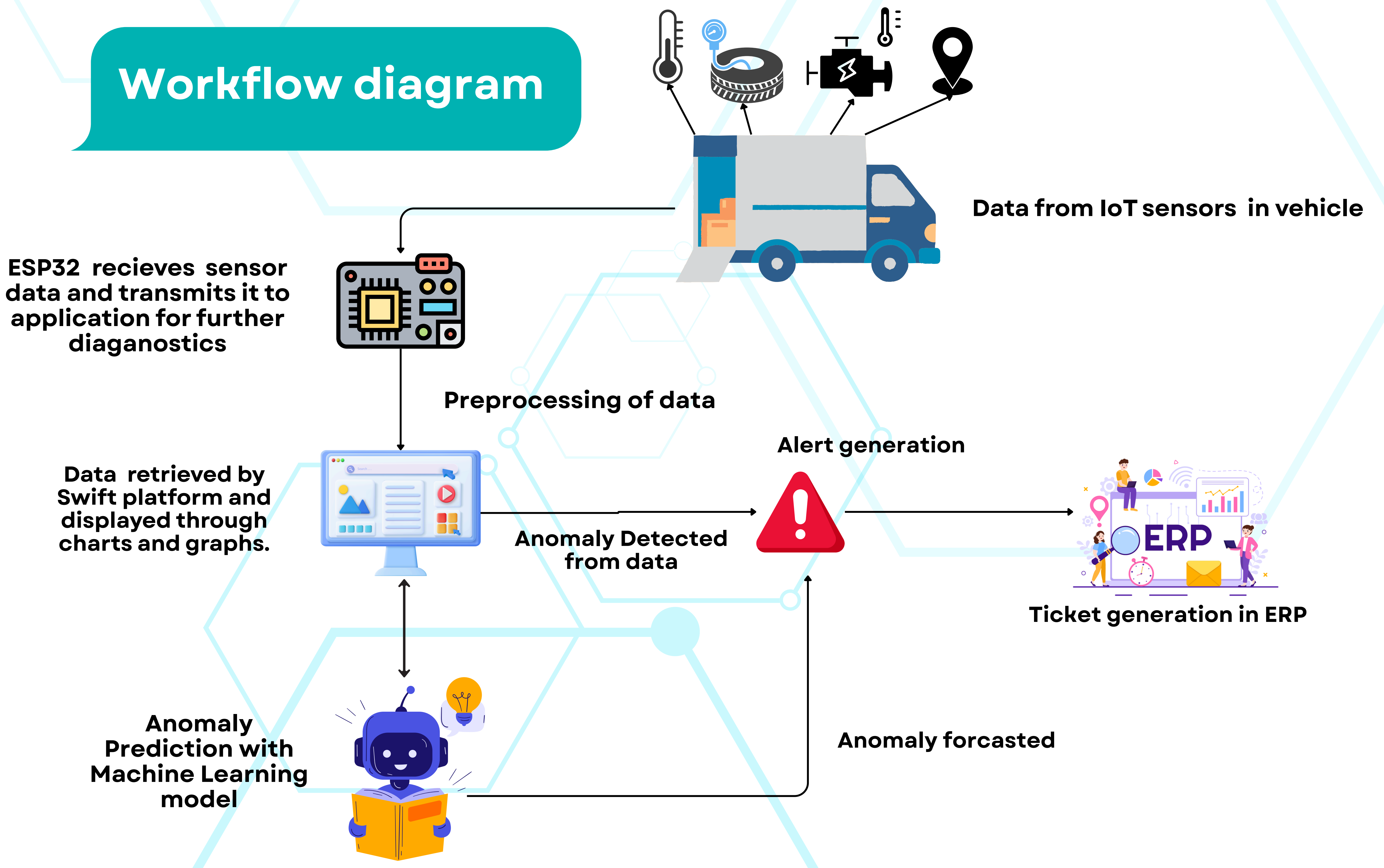
This system uses IoT, GPS, and machine learning to enable real-time tracking and predictive maintenance of flight catering vehicles via the Swift Platform. It reduces downtime, cuts costs, and ensures timely service through automated alerts, ERP ticketing, and a live dashboard.

ENGINE
START

Theory/Methods

- Vehicle data captured via sensors and GPS
- ESP32 sends data to central web server
- Web server processes and stores data
- Swift platform retrieves and analyzes data
- ML model checks for predictive maintenance needs
- Maintenance requirement is evaluated
- ERP ticket generated if maintenance is needed
- Live vehicle tracking shown via GPS on dashboard

Workflow diagram



Data from IoT sensors in vehicle

ESP32 recieves sensor data and transmits it to application for further diaganostics

Preprocessing of data

Data retrieved by Swift platform and displayed through charts and graphs.

Anomaly Detected from data

Alert generation

Ticket generation in ERP

Anomaly Prediction with Machine Learning model

Anomaly forcasted

Tech Stack

mongoDB

salesforce

swift

ESP32

DigitalOcean

ubuntu

Future Enhancements

- Integrate real-time traffic data for better maintenance predictions
- Include terrain conditions and driver behavior in analysis
- Leverage advanced AI for smarter predictive models
- Use IoT advancements for more scalable and dynamic systems
- Develop tailored solutions for different operational needs

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